

SALES PERFORMANCE ANALYSIS

Root Cause Investigation of Revenue Decline (2011-2012)

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Executive Summary

This project investigates a significant 43.1% year-over-year decline in revenue between 2011 and 2012 for a retail business using an end-to-end analytics approach. The primary objective was to identify the key business drivers behind the decline and determine where the business should focus future investigation and corrective actions.

The analysis combined Excel, SQL Server, and Power BI to complete the full analytics lifecycle. Excel was used for data profiling and preparation, SQL Server was used for data validation, Exploratory Data Analysis, and root cause investigation, and Power BI was used to build a semantic data model and an interactive dashboard for reporting.

A structured, hypothesis-driven investigation was conducted, beginning with validation of the reported revenue decline. Revenue was then decomposed into its two primary drivers—Average Selling Price (ASP) and Quantity Sold. The analysis showed that ASP decreased by only 2.2%, while Quantity Sold declined by 41.9%, confirming that the reduction in sales volume was the primary driver of the revenue decline.

Subsequent drill-down analysis revealed that the decline was concentrated within specific business segments rather than affecting the entire business uniformly. The largest reductions occurred during April, May, June, July, and October, with the Regular product class contributing most of the sales volume decline. Five product categories and nine product subcategories accounted for most lost sales, while geographically, the decline was heavily concentrated in North America, particularly the United States. Product-level and store-level analyses showed that the decline was broadly distributed across many products and stores rather than being caused by a small number of underperforming products or locations.

The investigation concludes that the revenue decline was primarily driven by lower sales volume concentrated within specific products, time periods, and geographic markets. However, the available dataset identifies where the decline occurred rather than why it occurred. Additional business data—such as inventory availability, marketing activities, pricing strategy, sales operations, customer behavior, and external market conditions—would be required to determine the underlying operational and strategic causes. The findings provide a focused direction for future business investigation while demonstrating a complete end-to-end analytical workflow for identifying and communicating business insights.

Project Overview

1. Project Objective

The objective of this project is to identify the primary drivers behind the revenue decline by applying an end-to-end analytics workflow using Excel, SQL Server, and Power BI. The project combines data preparation, validation, root cause analysis, and dashboard development to generate data-driven business insights.

2. Dataset Overview

The analysis is based on an open-source retail dataset obtained from Kaggle. The dataset consists of seven related tables that capture sales transactions, product hierarchy, store information, geography, and calendar data. Together, these tables support end-to-end sales analysis by enabling investigation across multiple business dimensions.

The table below summarizes the structure of each table, including its row count, column count, grain, primary key, and foreign key relationships.

Table	Rows	Columns	Grain
Sales	69999	5	One sale
Product	1690	9	One product
Product Subcategory	45	3	One product subcategory
Product Category	9	2	One product category
Store	307	5	One store
Geography	675	4	One geography
Calendar	732	8	One date

Figure 1: Dataset Overview

A detailed description of each table and column is provided in the project's data dictionary available in the GitHub repository.

3. Assumptions

Since this project uses a publicly available sample dataset and the available business information is limited, the following assumptions were made before conducting the analysis:

1. Sales transactions are assumed to be complete for the available reporting period.

2. Although the Calendar table contains dates for the entire year, sales transactions are available only between April and October. The analysis is therefore limited to this period.
3. Revenue is calculated as Unit Price × Quantity Sold, as information related to taxes, discounts, returns, and promotional pricing is not available in the dataset.
4. Product prices are assumed to be accurate because no invalid pricing records were identified during data validation.
5. The underlying business model (B2B or B2C) cannot be determined from the available data. Therefore, business interpretations and recommendations are limited to observable evidence.
6. Store names and brand names appear to be related; however, the dataset does not explicitly define this relationship. Consequently, brand-level analysis was considered outside the scope of this investigation.

Methodology

1. Overview

This project follows an end-to-end analytics workflow using Excel, SQL Server, and Power BI. Excel was used for data profiling and preparation, SQL Server for data validation, exploratory data analysis (EDA), and root cause investigation, while Power BI was used to develop the semantic data model, create DAX measures, and build interactive dashboards for communicating business insights.

SQL queries, analytical views, and Common Table Expressions (CTEs) were used to support data validation, exploratory analysis, and root cause investigation.

The investigation followed a hypothesis-driven approach. Beginning with validation of the reported revenue decline, the analysis progressively examined the key business drivers before drilling down across time, product, geography, and store dimensions to identify the primary factors contributing to the decline in sales performance.

2. Data Preparation

The raw dataset was imported into Power Query Editor for data profiling and cleaning before being loaded into SQL Server for analysis. Each table was reviewed to verify data types, assess missing values, identify duplicate records, examine potential outliers, and validate overall data consistency. As the dataset was already well maintained, only minor cleaning and standardization tasks were required before analysis.

3. Data Preparation Summary

Table	Data Preparation Performed
Sales	Verified data types, missing values, duplicate records, and business-critical fields. Converted the Date_Key column from Date/Time to Date . No data quality issues were identified.
Product	Standardized category names by replacing "and" with "&".
Geography	Confirmed that three blank values in the Country column represented continent-level records where country value was not applicable. No data modifications were required.
Product Category	No data quality issues identified
Product	
Subcategory	
Store	
Cale	

Figure 2: Data Preparation Summary

4. Data Validation

Following data preparation, the cleaned tables were imported into SQL Server to build a relational database for analysis. A snowflake schema was implemented to preserve the hierarchical relationships between dimensions while minimizing data redundancy.

- Schema Design & Implementation
- Imported the cleaned tables into SQL Server.
- Assigned primary keys to uniquely identify records.
- Defined foreign key constraints to establish relationships between the fact and dimension tables and enforce referential integrity.

5. Data Validation Summary

After the database was created, several validation checks were performed to ensure data integrity and readiness for analysis.

- Verified row counts to confirm that all records were imported successfully.
- Checked for invalid numeric values in business-critical fields, including Sales Quantity, Unit Price, and Unit Cost.
- Calculated additional business metrics, including Revenue, Cost, Profit, and Profit Margin, to support subsequent analysis.

6. SQL Investigation & Root Cause Analysis

To support efficient business investigation, two analytical SQL views were created to simplify data retrieval and reduce query complexity throughout the analysis.

- **vw_Yearly_Metrics** aggregates yearly business metrics, including Total Revenue, Total Quantity Sold, and Average Selling Price (ASP), providing a high-level view of company performance.
- **vw_Sales_Analysis** combines the Sales fact table with the Product, Product Subcategory, Product Category, Store, Geography, and Calendar dimensions to create a business-friendly dataset for multidimensional analysis.

The SQL investigation was performed using joins, aggregate functions, window functions, Common Table Expressions (CTEs), and analytical views to efficiently validate hypotheses and perform progressive drill-down analysis. Reusable SQL views reduced code duplication and simplified querying, while CTEs improved query readability and maintainability during complex investigations.

The investigation followed a hypothesis-driven approach. It began by validating the reported revenue decline before examining its two primary drivers: Average Selling Price (ASP) and Quantity Sold. Since the change in ASP was relatively small compared with the decline in Quantity Sold, subsequent analysis focused on sales volume across time, product hierarchy, geography, and store dimensions to identify the primary contributors to the revenue decline.

Project Workflow (Technical pipeline)

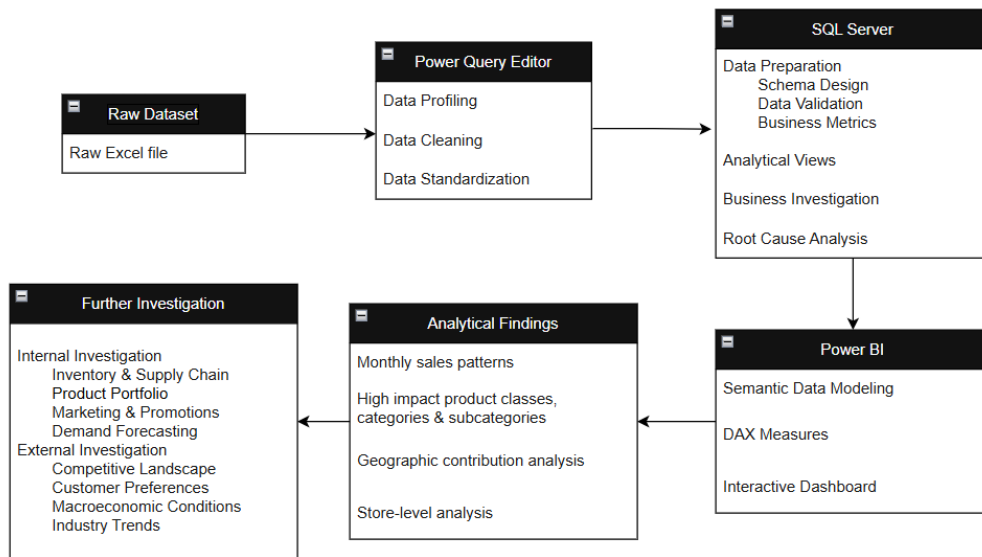


Figure 3: Project Workflow

Investigation Workflow (Business investigation)

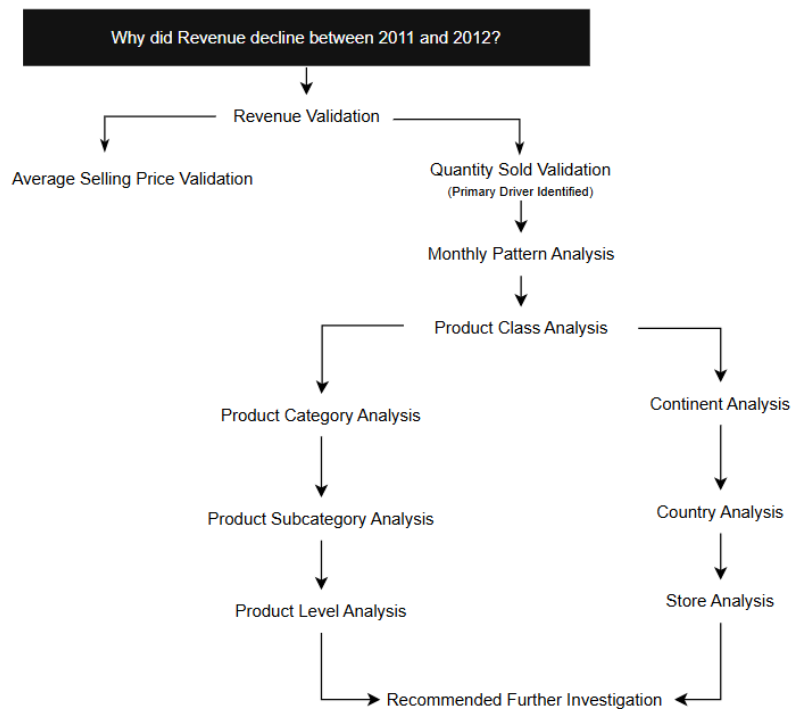


Figure 4: Investigation Workflow

Exploratory Data Analysis & Findings

1. Revenue Validation

H1: Has revenue declined by approximately 43.1% between 2011 and 2012?

- **Key Result**

- a. Revenue (2011): \$152.2M
- b. Revenue (2012): \$86.6M
- c. Revenue Change: -\$65.6M
- d. YoY Change: -43.1%

- **Finding**

The analysis confirms that total revenue declined by **43.1%** between 2011 and 2012, validating the reported business problem. Since the decline is substantial, further investigation is required to determine the underlying business drivers.

2. Average Selling Price Validation

H2: Did a decrease in Average Selling Price (ASP) cause the revenue decline?

- **Key Result**

- a. Average Selling Price (2011): \$343.82
- b. Average Selling Price (2012): \$336.39
- c. Average Selling Price Change: -\$7.43
- d. YoY Change: -2.2%

- **Finding**

Average Selling Price decreased by **\$7.43 (-2.2%)** between 2011 and 2012. Compared with the **43.1% decline in revenue**, this change is relatively small, indicating that pricing had a minimal impact on the revenue decline. Therefore, **Average Selling Price is not the primary driver**, and the investigation should focus on **Quantity Sold**.

3. Quantity Sold Investigation

H3: Did the decline in Quantity Sold contribute to the overall revenue decline?

- **Key Result**

- a. Quantity Sold (2011): 442.7K units
- b. Quantity Sold (2012): 257.3K units
- c. Quantity Sold Change: -185.3K units

d. YoY Change: -41.9%

- **Finding**

Quantity Sold declined by **185.3K units (-41.9%)** between 2011 and 2012. This decline closely aligns with the **43.1% decrease in revenue**, while Average Selling Price decreased by only **2.2%**. These findings indicate that the reduction in **sales volume** was the primary driver of the revenue decline. Therefore, the remainder of the investigation focuses on identifying **which months, product segments, geographies, and stores contributed most to the decrease in Quantity Sold.**

4. Monthly Patterns

H4: Did Quantity Sold decline throughout the year, or was there a seasonal pattern?

- **Key Result**

- The decline was concentrated in April, May, June, July, and October.
- These five months accounted for 102.08% of the total decline in Quantity Sold.
- May (-51.1K units) and June (-50.9K units) recorded the largest decreases.
- August (+15.8%) and September (+19.2%) were the only months that showed positive YoY growth.

- **Finding**

The decline in Quantity Sold was **not evenly distributed throughout the year**. Instead, it was concentrated in **April, May, June, July, and October**, with May and June experiencing the largest reductions. In contrast, August and September recorded growth, partially offsetting the overall decline. These findings suggest that the revenue decline was driven by **specific periods of weak sales performance rather than a consistent year-round decrease**. Therefore, the investigation continues by identifying **which product segments contributed most to the decline during these affected months.**

5. Product Class Analysis

H5: Which Product Class contributed most to the decline in Quantity Sold during the affected months?

- **Key Result**

- Regular product class contributed 103.02% of the total decline in Quantity Sold.
- Deluxe product-class contributed an additional 14.71% of the decline.
- Economy product class recorded an increase of 33.6K units (+2178.6%), partially offsetting the losses from the other product classes.

- **Finding**

Although both **Regular** and **Deluxe** product classes experienced substantial declines, the **Regular** class alone accounted for **103.02%** of the total reduction in Quantity Sold, making it the dominant contributor to the revenue decline. Since the objective of this investigation is to identify the primary root cause, the subsequent analysis focuses on the **Regular** product class to determine which product categories, subcategories, and individual products were responsible for the decline.

6. Product Category Analysis

H6: Which product categories within the Regular product class contributed most to the decline in Quantity Sold?

- **Key Result**

- Cameras & Camcorders contributed 29.8% of the total decline.
- Cell Phones contributed 24.6%.
- Computers contributed 21.1%.
- TV & Video contributed 14.1%.
- Music, Movies & Audio Books contributed 9.3%.
- Together, these five product categories accounted for 98.9% of the total decline in Quantity Sold.
- Audio contributed only 1.2% and was excluded from further investigation.

- **Finding**

The decline in Quantity Sold was highly concentrated within five product categories, which together explained **98.9% of the total reduction** in sales volume. Among them, **Cameras & Camcorders and Cell Phones were the largest contributors, followed by Computers, TV & Video, and Music, Movies & Audio Books**. Since the Audio category contributed only 1.2% of the overall decline, it was considered immaterial and excluded from further analysis. Therefore, the investigation proceeds by examining the product subcategories within the five major categories to identify the specific products driving the decline.

7. Product Subcategory Analysis

H7: Which Product Subcategories had a highest negative growth in affected months, product class and categories?

- **Key Result**

- 9 product subcategories accounted for 86.7% of the total decline in Quantity Sold.
- The largest contributors were Digital Cameras (14.9%), Digital SLR Cameras (10.4%), Printers, Scanners & Fax (10.1%), Projectors & Screens (9.4%), Movie DVD (9.4%), Touch Screen Phones (9.2%), Home & Office Phones (7.9%), Smart Phones & PDAs (7.8%), and Home Theater System (7.6%), together accounting for 86.7% of the total decline.
- Laptops** were the only subcategory showing growth (+63.7%), partially offsetting the overall decline.

- **Finding**

The decline in Quantity Sold was concentrated within **nine product subcategories**, which together explained **86.7%** of the overall reduction in sales volume. These subcategories span the previously identified high-impact product categories, indicating that the revenue decline was driven by a relatively small group of products rather than being evenly distributed across the entire product portfolio. Although the **Laptops** subcategory recorded strong growth, it was insufficient to offset the losses from the other subcategories. Therefore, the investigation proceeds with **product-level analysis** to identify the individual products responsible for the decline.

8. Product-level Analysis

H8: Which Products had a highest negative growth in affected months, product class, categories & subcategories?

- **Key Result**

- The decline was distributed across multiple products rather than being driven by a few individual products.
- Only a small number of products recorded growth, partially offsetting the overall decline.

- **Finding**

The reduction in Quantity Sold was **broadly spread across many individual products**, indicating that the decline was not caused by a small number of underperforming products. Therefore, no further product-level investigation was

conducted, and the analysis concludes that the primary drivers were identified at the **product category and subcategory levels**.

9. Continent Analysis

H9: Which continents experienced the largest decline in Quantity Sold in affected months, and product segments?

- **Key Result**

- c. North America contributed 85.96% of the total decline in Quantity Sold.
- d. Europe contributed 13.81%.
- e. Asia contributed only 0.23%.

- **Finding**

The decline in Quantity Sold was **heavily concentrated in North America**, which accounted for **85.96%** of the total reduction in sales volume. Therefore, the investigation proceeds to the **country level** to identify the specific markets driving the decline.

10. Country Analysis

H10: Which country in North America contributed the largest decline in Quantity Sold in affected months and product segments?

- **Key Result**

- a. United States contributed 94.28% of the total decline in Quantity Sold.
- b. Canada contributed 5.72%.

- **Finding**

The decline in Quantity Sold was overwhelmingly concentrated in the United States, accounting for 94.28% of the total reduction in sales volume. Therefore, any further business investigation or corrective actions was primarily focus on the U.S. market.

11. Store-level Analysis

H11: Which stores contributed highest in the decline of Quantity Sold?

- **Finding**

No individual store accounted for a disproportionately large decline in Quantity Sold. Instead, the decline was distributed across stores within the identified months, product segments, and geographic regions, indicating that improvement efforts should target the broader business rather than specific stores.

Most of the affected products belong to the **Contoso** brand. However, since the dataset does not provide sufficient information to evaluate brand performance, **brand-level analysis was outside the scope of this investigation.**

Business interpretation

The investigation confirms that the revenue decline was **not caused by pricing**. Average Selling Price decreased by only 2.2%, while Quantity Sold declined by 41.9%. This indicates that the primary business issue is a reduction in sales volume rather than pricing strategy.

The decline is **not companywide**. It is concentrated in:

1. Specific months (April, May, June, July, and October)
2. The Regular product class
3. Five major product categories
4. Nine key product subcategories
5. Primarily the United States

This shows that the problem is focused on business segments rather than affecting all products, markets, and periods equally.

Because the decline is concentrated in specific products and markets, it is unlikely to be caused by random variation. Instead, it suggests that underlying operational or market factors—such as inventory availability, marketing effectiveness, customer demand, or competitive pressure—may have influenced sales performance.

The available dataset identifies **where** the decline occurred but does not explain **why** it occurred. Additional business data would be required to determine the root operational or market causes.

Note: The Store dimension contains a single status value per store and does not track status over time. Stores marked as Off were found to have valid sales revenue during the analysis period, indicating that the field represents the current store status rather than historical operational status. Therefore, no filtering was applied based on Store Status to avoid excluding valid historical transactions.

Dashboard Development

1. Dashboard Design Approach

The dashboard was designed to present the investigation findings in a logical, business-focused sequence. Each page progressively narrows the analysis from the overall business problem to the specific drivers responsible for the revenue decline.

The investigation follows a three-level investigation approach:

1. Executive Overview
2. Product Performance
3. Geography Performance

This structure enables stakeholders to understand both the overall business impact and the underlying factors driving the performance decline.

2. KPIs

- Revenue KPI displays the total revenue for 2012, the Year-over-Year (YoY) percentage change, and a mini comparison chart showing each year's contribution to total revenue.
- Quantity Sold KPI presents the total Quantity Sold for 2012, the YoY percentage change, and a mini comparison chart illustrating each year's contribution to total Quantity Sold.
- Average Selling Price (ASP) KPI displays the average selling price for 2012, the YoY percentage change, and a mini comparison chart comparing ASP across both years.

3. Visual Selection Rationale

The dashboard uses visuals that directly support business investigation and highlight the key drivers of the revenue decline.

- 1 KPI Cards summarize overall business performance using Revenue, Quantity Sold, and Average Selling Price (ASP).
- 2 Clustered Column & Line Charts compare business performance across years while highlighting Year-over-Year (YoY) changes.
- 3 Pareto Charts identify the product categories, subcategories, and geographic regions contributing most to the overall Quantity Sold decline.
- 4 Tables provide detailed rankings and support deeper analysis of high-impact product subcategories and geographic locations.

4. Filters Applied

To ensure consistency with the investigation scope, page-level filters were applied where appropriate.

- **Executive Overview**

- No page-level filters were applied.
- The dashboard presents the overall business performance for 2011 and 2012 across all products, months, and geographies.
- **Product Performance**
 - Filters applied on Product Class (Regular) and Months (April, May, June, July & October)
 - These filters focus the analysis on the product class and months identified during the SQL investigation as the primary contributors to the Quantity Sold decline.
- **Geography Performance**
 - Filters applied for Product Class (Regular), Months (April, May, June, July & October) , Product Category (Cameras & Camcorders, Cell Phones, Computers, TV & Video, Music, Movies & Audio Books), and Product Subcategory (Digital Cameras, Digital SLR Cameras, Printers, Scanners & Fax, Projectors & Screens, Movie DVD, Touch Screen Phones, Home & Office Phones, Smart Phones & PDAs, and Home Theater System)

These filters narrow the analysis to the highest-impact product segments, enabling the dashboard to identify the geographic regions responsible for most of the Quantity Sold decline.

Dashboard Preview

1. Executive Overview

The Executive Overview provides a high-level summary of the business performance between 2011 and 2012. It presents the overall changes in Revenue, Quantity Sold, and Average Selling Price (ASP), enabling stakeholders to quickly assess the extent of the decline.

The dashboard then investigates the primary drivers of the revenue decline by analyzing monthly sales patterns and product class performance. Quantity Sold trends, Year-over-Year (YoY) changes, and percentage contribution analyses highlight the periods and product classes that contributed most significantly to the overall decrease.

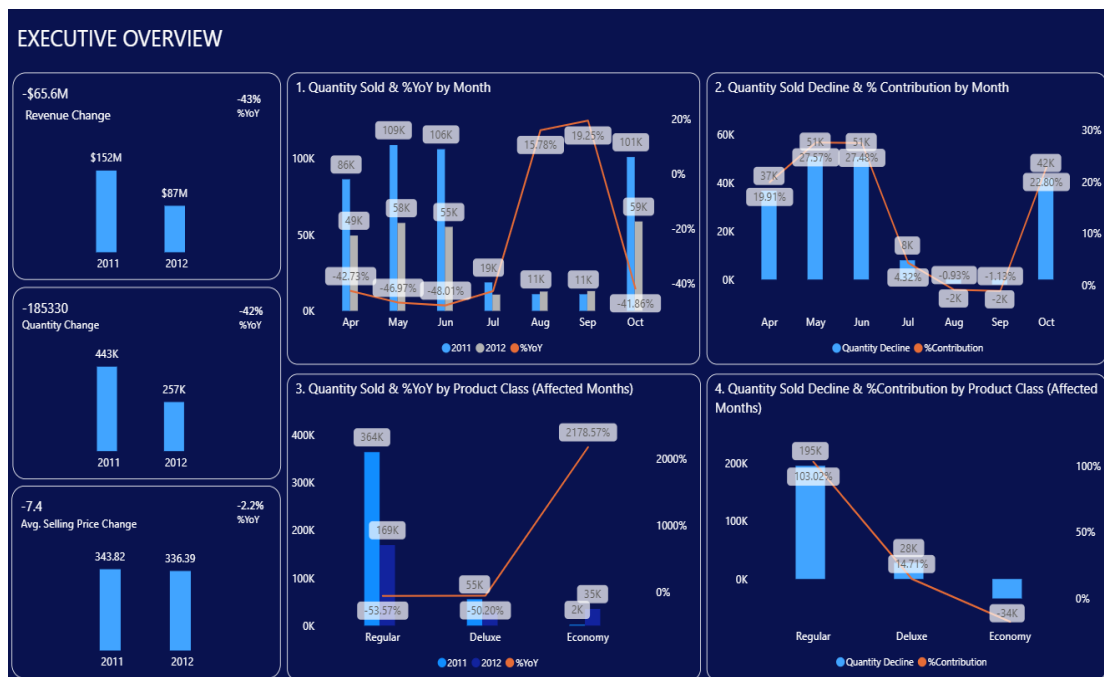


Figure 5: Dashboard 1 - Executive Overview

2. Product Performance

This page focuses on identifying the product segments responsible for the decline in Quantity Sold. The analysis begins at the Product Category level before drilling down to Product Subcategory, allowing high-impact product groups to be identified.

The dashboard combines Quantity Sold comparisons, Year-over-Year (YoY) performance, percentage contribution analysis, and detailed subcategory rankings to determine which product segments contributed most to the overall business decline.

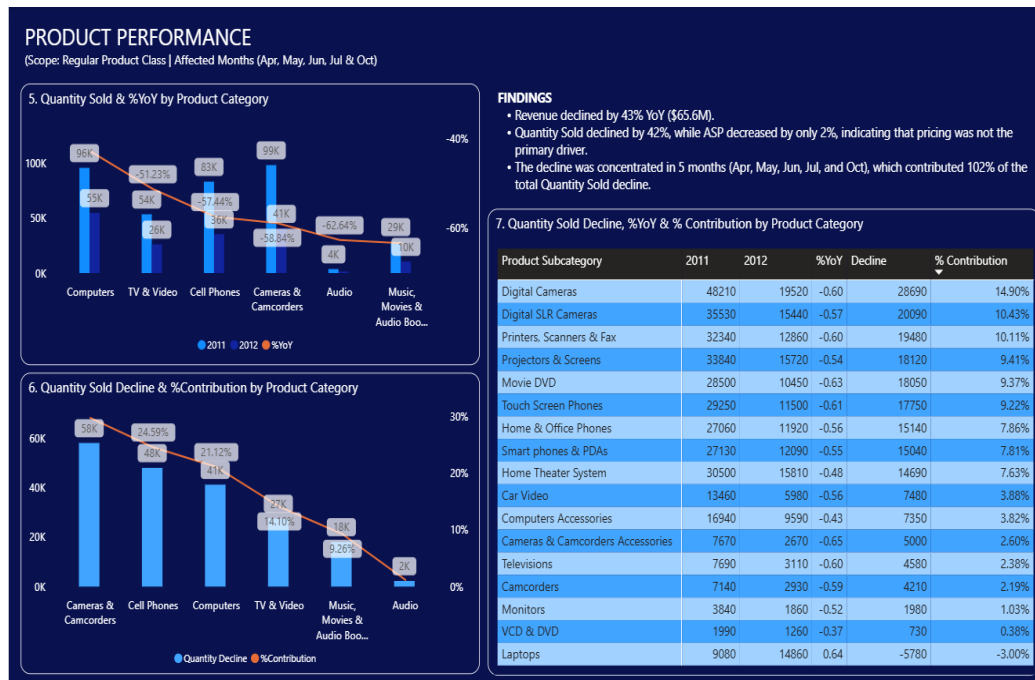


Figure 6: Dashboard 2 - Product Performance

3. Geography Performance

The Geography Performance dashboard investigates the regional distribution of the Quantity Sold decline. The analysis first compares performance across continents before drilling down to the country level within the most affected region.

Contribution analysis highlights the geographic concentration of the decline and identifies the primary markets responsible for the reduction in sales volume. This enables stakeholders to prioritize recovery efforts toward the regions with the greatest business impact.

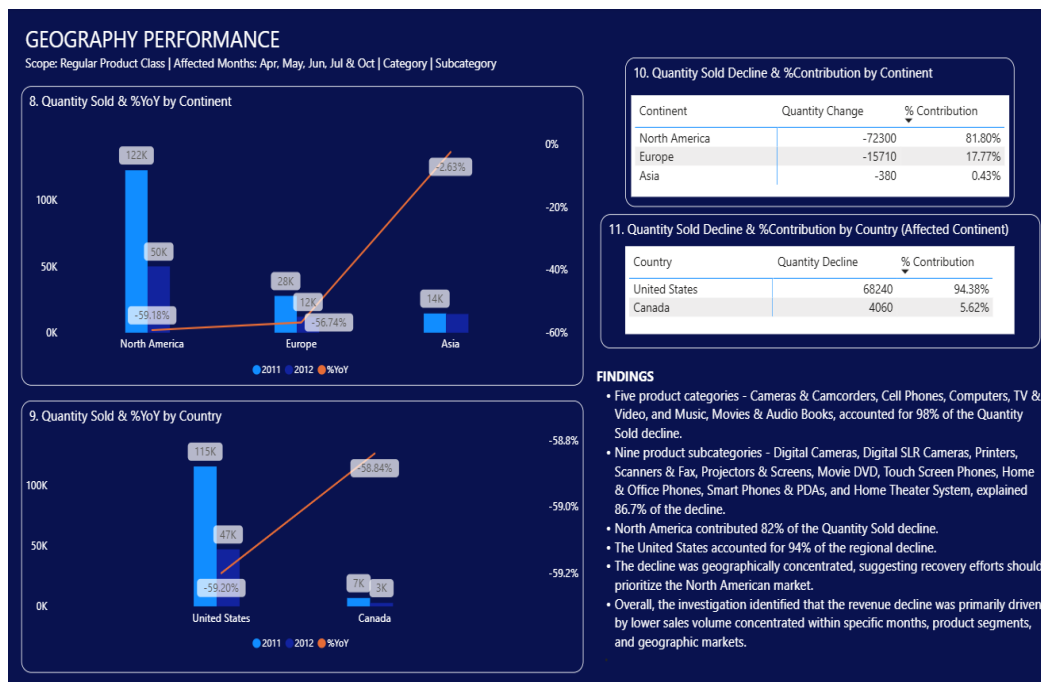


Figure 7: Dashboard 3 - Geography Performance

Further Investigation Areas

Although the analysis successfully identified where the revenue decline occurred, the available dataset does not explain why it occurred. Additional operational and business data would be required to investigate the underlying causes. The following areas represent potential directions for future analysis.

1. Internal Business Factors

- **Inventory & Supply Chain**

- Objective: Determine whether inventory availability or supply chain disruptions contributed to the decline in sales.
- Potential Hypothesis
 - Were the effected products out of stock during the months with the highest sales decline?
 - Were stockouts concentrated in the US?
 - Were suppliers leading time longer than expected?
 - Were there warehouses or distribution delays affecting product availability?
- Additional Data Required
 - Inventory Levels
 - Stockout Records

- iii. Supplier Lead Times
- iv. Warehouse Distribution Data

- **Marketing & Promotions**

- d. Objective: Evaluate whether changes in marketing strategy or promotional activities contributed to lower sales.
- e. Potential Hypothesis
 - i. Was marketing campaigns reduced or discontinued?
 - ii. Was there any major change in the strategy over the last one year?
 - iii. Were promotional campaigns less effective?
 - iv. Was advertising spend reduced than the previous year?
- f. Additional Data Required
 - i. Marketing Campaign Data
 - ii. Advertising Spend
 - iii. Promotion Calendar

- **Sales Operations**

- g. Objective: Determine whether pricing decisions influenced customer purchasing behavior despite the relatively small change in ASP.
- h. Potential Hypothesis
 - i. Did sales team performance decline?
 - ii. Were there staffing or territory changes?
 - iii. Were sales incentives modified?
- i. Additional Data Required
 - i. Store Team Performance
 - ii. Store Operations Data
 - iii. Territory Assignments

- **Customer Behavior**

- j. Objective: Determine whether changes in customer purchasing behavior contributed to the decline in sales performance.
- k. Potential Hypothesis
 - i. Did customer acquisition decline between 2011 and 2012?
 - ii. Did customer churn increase during the affected period?
 - iii. Did customers shift their purchases to other product categories?
 - iv. Did customer satisfaction decline, leading to reduced repeat purchases?

I. Data Required

- i. Customer transaction history
- ii. Customer Acquisition and churn data
- iii. Loyalty program Data
- iv. Customer Feedback and surveys

2. External Business Factors

• Market Demand

a. Objective

Determine whether changes in overall market demand or external economic conditions contributed to the decline in sales.

b. Potential Hypothesis

- i. Did overall demand for the affected product categories decline between 2011 and 2012?
- ii. Did competitors gain market share during the same period?
- iii. Were there changes in consumer preferences that reduced demand for the affected products?
- iv. Did economic conditions influence customer purchasing behavior, particularly in the United States?

c. Additional Data Required

- i. Industry sales and market reports
- ii. Competitor sales or market share data
- iii. Consumer market research
- iv. Economic indicators (e.g., consumer confidence, inflation, unemployment)

Conclusion

This investigation successfully identified the business areas where the revenue decline occurred and established that the primary driver was a reduction in sales volume rather than pricing. However, the available dataset was limited to transactional sales data and did not include operational, marketing, customer, or external market information required to determine the underlying business causes.

The proposed investigation areas provide a structured roadmap for future analysis by identifying additional data sources and business hypotheses that can be explored. Incorporating these

datasets would enable a more comprehensive understanding of the operational and market factors influencing sales performance and support more targeted business decisions.